WORKED SOLUTION

STEP II 2019, Question 3

Pure • Bounds on Polynomial Roots

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Read the question before you go on. It is in the STEP II paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

The key idea

The triangle inequality lets us bound a polynomial by replacing every term with its size, and the sizes form a geometric series when $|x| < 1$. That gives a lower bound on small roots. For large roots, the trick is to look at $1/\omega$: reversing the coefficients of f gives a polynomial of exactly the same shape, so the small-root result does the work a second time. In part (ii) the bounds cut the search for integer roots down to four candidates.

Solution

The triangle inequality

Both sides are non-negative, so it is enough to compare their squares. Since $x_1x_2 \leq |x_1x_2| = |x_1||x_2|$,

$$|x_1 + x_2|^2 = x_1^2 + 2x_1x_2 + x_2^2 \leq |x_1|^2 + 2|x_1||x_2| + |x_2|^2 = (|x_1| + |x_2|)^2,$$

and taking non-negative square roots gives $|x_1 + x_2| \leq |x_1| + |x_2|$.

For n numbers we use induction on n. The case $n = 1$ is an equality and $n = 2$ is done. If the result holds for $n - 1$ numbers, then applying the two-number case to $x_1 + \cdots + x_{n-1}$ and x_n,

$$\begin{aligned} |x_1 + \cdots + x_{n-1} + x_n| &\leq |x_1 + \cdots + x_{n-1}| + |x_n| \\ &\leq |x_1| + \cdots + |x_{n-1}| + |x_n|, \end{aligned}$$

which is the result for n numbers.

Part (i)(a)

We have $\mathrm{f}(x) - 1 = a_1 x + a_2 x^2 + \cdots + a_{n-1}x^{n-1} + x^n$. By the triangle inequality, then $|a_i| \leq A$, and then $1 \leq A$ for the last term,

$$\begin{aligned} |\mathrm{f}(x) - 1| &\leq |a_1||x| + \cdots + |a_{n-1}||x|^{n-1} + |x|^n \\ &\leq A(|x| + |x|^2 + \cdots + |x|^n). \end{aligned}$$

Because $0 \leq |x| < 1$, the finite geometric sum is at most the infinite one, all of whose terms are non-negative:

$$|x| + |x|^2 + \cdots + |x|^n \leq \sum_{k=1}^{\infty} |x|^k = \frac{|x|}{1 - |x|}.$$

Hence $|\mathrm{f}(x) - 1| \leq \dfrac{A|x|}{1 - |x|}$, as required.

Part (i)(b)

Since $\mathrm{f}(\omega) = 0$, we have $|\mathrm{f}(\omega) - 1| = 1$, so part (a) gives

$$1 \leq \frac{A|\omega|}{1 - |\omega|}.$$

As $1 - |\omega| > 0$ we may multiply through without changing the direction: $1 - |\omega| \leq A|\omega|$, that is $1 \leq (1 + A)|\omega|$, so

$$|\omega| \geq \frac{1}{1 + A}.$$

The upper bound is immediate: $|\omega| < 1 \leq 1 + A$. So $(*)$ holds.

Part (i)(c)

Now $|\omega| \geq 1$. The lower bound is easy: $|\omega| \geq 1 > \frac{1}{1+A}$, since $A \geq 1 > 0$. The upper bound is the real content.

If $|\omega| = 1$ then $|\omega| \leq 1 + A$ at once, so suppose $|\omega| > 1$. Put $\mathrm{g}(y) = y^n \mathrm{f}(1/y)$, which is

$$\mathrm{g}(y) = 1 + a_{n-1}y + a_{n-2}y^2 + \cdots + a_1 y^{n-1} + y^n.$$

? **Ask yourself** Why is g allowed to use the results of parts (a) and (b)?

It has exactly the form of f: constant term 1, leading coefficient 1, and its other coefficients are $a_{n-1}, \ldots, a_1$ in reverse order, each still of size at most A. So part (b) applies to g with the same A.

Since $\omega \neq 0$, $\mathrm{g}(1/\omega) = \omega^{-n}\mathrm{f}(\omega) = 0$, so $1/\omega$ is a real root of g with $|1/\omega| < 1$. Part (b) for g gives

$$\frac{1}{|\omega|} \geq \frac{1}{1 + A}, \qquad \text{so} \qquad |\omega| \leq 1 + A.$$

Therefore $(*)$ holds for every real root of f, whatever its size. (Note that 0 is never a root, since $\mathrm{f}(0) = 1$.)

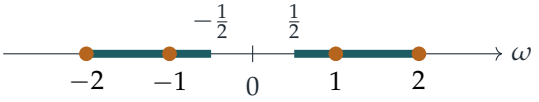
Part (ii)

Dividing by 135 and writing the terms in ascending powers, the equation is $\mathrm{f}(x) = 0$ with

$$\mathrm{f}(x) = 1 - \tfrac{126}{135}x - \tfrac{91}{135}x^2 - \tfrac{100}{135}x^3 - x^4 + x^5.$$

This has the form of part (i) with $n = 5$: constant term and leading coefficient both 1, and the other coefficients have sizes $\frac{126}{135}, \frac{91}{135}, \frac{100}{135}, 1$, all at most $A = 1$. So every real root satisfies

$$\tfrac{1}{2} \le |\omega| \le 2.$$



The only integers in the allowed region (shaded) are ± 1 and ± 2. Evaluating the original quintic $p(x) = 135x^5 - 135x^4 - 100x^3 - 91x^2 - 126x + 135$:

$$\begin{aligned} p(1) &= 135 - 135 - 100 - 91 - 126 + 135 = -182, \\ p(-1) &= -135 - 135 + 100 - 91 + 126 + 135 = 0, \\ p(2) &= 4320 - 2160 - 800 - 364 - 252 + 135 = 879, \\ p(-2) &= -4320 - 2160 + 800 - 364 + 252 + 135 = -5657. \end{aligned}$$

So the only integer root is $x = -1$. Indeed

$$p(x) = (x+1)(135x^4 - 270x^3 + 170x^2 - 261x + 135),$$

and the quartic factor is 971 at $x = -1$, so -1 is a simple root.

Checks

- Multiply out $(x+1)(135x^4 - 270x^3 + 170x^2 - 261x + 135)$ to recover the quintic.
- The quintic has two other real roots, about 0.595 and 1.725; both lie in $[\frac{1}{2}, 2]$, as part (i) predicts.
- Try $\mathrm{f}(x) = 1 - 2x + x^2 = (1-x)^2$ with $A = 2$: the root 1 lies in $[\frac{1}{3}, 3]$. The bounds were also confirmed by computer on hundreds of random polynomials.

Common slips

- Forgetting the x^n term in part (a): its coefficient is 1, not some a_n, and it is $A \ge 1$ that lets it join the geometric series.
- In part (c), trying to reuse part (a) directly at ω when $|\omega| \ge 1$: the geometric series no longer converges. Go to $1/\omega$ instead.
- Testing all the divisors of 135 as candidate roots in part (ii): legitimate, but the bounds from part (i) leave only four to try.

Beyond STEP

Bounding the roots of a polynomial in terms of its coefficients is a classical theme; this is a version of Cauchy's bound, and the same argument works word for word for complex roots, with modulus in place of absolute value. The reversal $y^n \mathrm{f}(1/y)$ swaps small roots with large ones and is a standard tool in algebra and complex analysis. For bounds like these and much more on the geometry of complex roots, try Needham, *Visual Complex Analysis*.

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